XML YAHOO! ANSWERS

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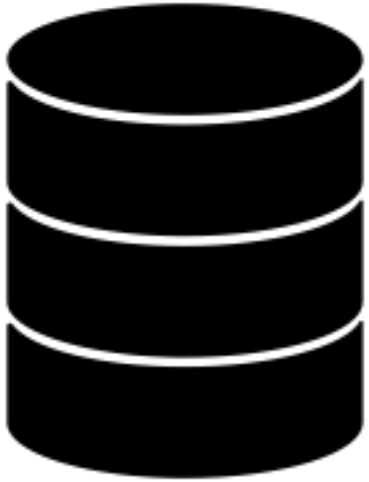
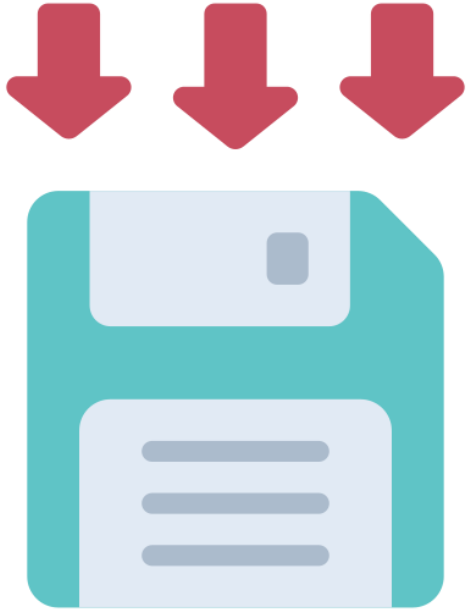
01

Introducción

Antecedente Proyecto.





02

Dataset

Exploración, Categorías y Entorno.

Exploración del *Dataset*

Nombre del Archivo	Descripción
extract_maincats.py	Extrae las 178 categorías principales (<maincat>) del XML y las guarda en un archivo de texto. Es el paso inicial.
extract_subcats.py	Extrae todas las subcategorías únicas (<subcat>) para mapear la jerarquía completa del dataset.
validate_extract.py	Audita y valida que la extracción inicial de las 178 categorías se haya realizado sin omisiones.
analisis_profundo.py	Filtra y recomienda las mejores combinaciones de categorías temáticas (descartando las ya usadas). Clave para la selección final.
analizar_distribucion_general.py	Verifica la distribución real procesando los 4.4M de registros, confirmando los conteos de las categorías elegidas y sus variantes.

Categoría	Ocurrencias	Porcentaje
Science & Mathematics	183.354	29,9%
Education & Reference	234.604	38,3%
Social Science	58.207	9,5%
Arts & Humanities	137.080	22,4%
TOTAL	613.245	13,68% del dataset total

Emurasoft EmEditor

BASEX

03

Extracción de Categorías

Archivos y Resultados.

Desarrollo del Script de Extracción

2

División en Archivos por Categoría

Categoría	Preguntas	Tamaño estimado
yahoo_science.xml	183.354	Aproximadamente 450 MB
yahoo_education.xml	234.604	Aproximadamente 500 MB
yahoo_social.xml	58.207	Aproximadamente 200 MB
yahoo_arts.xml	137.080	Aproximadamente 380 MB

dividir_por_categoria.py

5 - 10 Minutos

04

Base de Datos Documental

BaseX y Resultados.

Manage Databases

yahoo_answers

prueba_yahoo_answers

yahoo_answers

Information Backups

yahoo_answers

Database Properties

NAME: yahoo_answers
SIZE: 1238 MB
NODES: 28122631
DOCUMENTS: 4
BINARIES: 0
VALUES: 0
TIMESTAMP: 2026-03-19T22:17:29.052Z
UPTODATE: true

Resource Properties

INPUTPATH: C:/Users/manuc/OneDrive/Escritorio/INFORMATICA/4º DE INFORMÁTICA/SEGUNDO CUATRI/BBDD_AVANZADAS/FullOct2007/xml_categorias/
INPUTSIZE: 1550 MB
INPUTDATE: 2026-03-19T22:12:22.881Z

Indexes

TEXTINDEX: true
ATTRINDEX: true
TOKENINDEX: false
FTINDEX: false
TEXTINCLUDE:
ATTRINCLUDE:
TOKENINCLUDE:
FTINCLUDE:
LANGUAGE: English
STEMMING: false
CASESENS: false
DIACRITICS: false
STOPWORDS:
UPDINDEX: false
AUTOOPTIMIZE: false
MAXCATS: 100
MAXLEN: 96
SPLITSIZE: 0

Drop... Rename... Copy... Open

file2* [yahoo_answers] - BaseX 12.2

Database Editor View Visualization Options Help

1 Result

Find

o/INFORMATICA/4º DE INFORMÁT

*.xml, *.xq*

Find contents...

BaseX

Context: db:get("yahoo_answer... Editor

file2*

```
1 count(//vespaadd)
```

OK 1 : 18

yahoo_arts.xml
yahooAnswers
vespaadd

yahoo_education.xml
yahooAnswers

yahoo_science.xml
yahooAnswers

yahoo_social.xml
yahooAnswers

1 Result, 6 b Result

613245

All Total Time: 11.38 ms Info

Query:
count(//vespaadd)

Query Plan:
<QueryPlan compiled="true" updating="false">
<Itr type="xs:integer" size="1">613245</Itr>
</QueryPlan>

Timing:
- Parsing: 0.78 ms
- Compiling: 7.18 ms
- Optimizing: 3.13 ms
- Evaluating: 0.02 ms
- Printing: 0.27 ms
- Total Time: 11.38 ms

db:get("yahoo_answers", "yahoo_arts.xml")/yahooAnswers/vespaadd 136 MB

05

1º Implementación y Revisión

Verificación y Bloques

Bloque 1 - Verificación General

613.245 Registros ✓

Atributo type del elemento document contiene únicamente el valor "wisdom" en la totalidad del dataset.

Bloque 2 - Verificación de Campos Supuestamente Obligatorios

Campo	Ausentes según XQuery	Decisión para el XSD
uri	0	Obligatorio - Confirmado.
subject	0	Obligatorio - Confirmado.
date	0	Obligatorio - Confirmado.
bestanswer	3.857 (0,63% del total)	Anomalía - modelar como opcional.
best_id	3.864 (0,63% del total)	Anomalía - modelar como opcional.
id	38	Anomalía - modelar como opcional.
nbestanswers	1	Anomalía - modelar como opcional.
lastanswerts	1	Anomalía - modelar como opcional.

Bloque 3 - Verificación de Campos Opcionales

Campo	Registros sin el campo	Porcentaje sobre 613.245
content	298.411	48,7%
vot_date	206.652	33,7%
res_date	3.819	0,62%
cat, maincat, subcat	0	0% - todos presentes en el subconjunto filtrado
qlang, qintl, language	0	0% - todos presentes en el subconjunto filtrado

Bloque 4 - Verificación de Tipos y Formatos

Campo(s)	Formato Esperado Inicial	Excepciones / Hallazgos	Formato Final / Patrón XSD
uri	Entero positivo	0	Entero positivo
id, best_id	u[0-9]+	0	u[0-9]+
Fechas (4 campos)	Entero no negativo	0	Entero no negativo
qintl	[a-z]{2}(ISO 3166-1 alpha-2)	4.757 registros (0,78%) con el valor "e1"	[a-z0-9]{2}
language	[a-z]{2}-[a-z]{2}	4.757 registros con el valor "es-e1"	[a-z]{2}-[a-z0-9]{2}

Bloque 5 - Análisis de nbestanswers

Métrica / Parámetro	Valor	Implicación Técnica / Hallazgo
Mínimo Observado	0	Valida la propiedad minOccurs="0" en el esquema XSD.
Máximo Observado	1.422	Indica una altísima variabilidad y picos de actividad en preguntas populares.
Media (Promedio)	5,77	Refleja el volumen estándar de participación por registro.
Nº de Registros sin Respuestas	1	Caso borde confirmado que justifica la opcionalidad del elemento.
Naturaleza del Dato	Alta dispersión	Comportamiento típico de plataformas de contenido generado por usuarios (UGC).

Bloque 6 - Campos Presentes pero Vacíos

Campo de Texto	Valores Vacíos Detectados	Restricción XSD (minLength)	Tipo de Dato Propuesto
subject	0	1	NonEmptyStringType
content	0	1	NonEmptyStringType
bestanswer	0	1	NonEmptyStringType
answer_item	0	1	NonEmptyStringType

06

Esquema XSD

XSD Original y XSD Propuesto

Esquema XSD Original

Campo	Obligatorio / Opcional	Justificación
uri	Obligatorio	0 ausentes en XQuery.
subject	Obligatorio	0 ausentes en XQuery.
date	Obligatorio	0 ausentes en XQuery.
bestanswer	Opcional (anomalía)	3.857 ausentes - XQuery contradice documentación oficial.
best_id	Opcional (anomalía)	3.864 ausentes - XQuery contradice documentación oficial.
id	Opcional (anomalía)	38 ausentes - XQuery contradice documentación oficial.
nbestanswers	Opcional (anomalía)	1 ausente - XQuery contradice documentación oficial.
lastanswerts	Opcional (anomalía)	1 ausente - XQuery contradice documentación oficial.
content	Opcional	298.411 ausentes (48,7%) - documentación oficial y XQuery.
vot_date	Opcional	206.652 ausentes (33,7%) - documentación oficial y XQuery.
res_date	Opcional	3.819 ausentes - documentación oficial y XQuery.
cat, maincat, subcat	Opcional	Doc oficial WebScope L6-1.
qlang, qintl, language	Opcional	Doc oficial WebScope L6-1.

Esquema XSD Propuesto

Tipo	Restricción introducida	Verificación XQuery que lo respalda
UnixTimestampType	xs:nonNegativeInteger	0 valores no numéricos en date, res_date, vot_date y lastanswerts
UserIdType	Patrón u[0-9]+	0 valores de id o best_id fuera del patrón cuando el campo está presente.
DocumentTypeEnum	Enumeración {wisdom}	distinct-values(@type) devuelve únicamente "wisdom" en 613.245 registros.
NonEmptyStringType	minLength="1"	0 campos de texto vacíos en subject, content, bestanswer ni answer_item.
LanguageCodeType	Patrón [a-z]{2}	Los 16 valores distintos de qlang son todos de 2 letras. 0 excepciones.
RegionCodeType	Patrón [a-z0-9]{2}	El valor "e1" en 4.757 registros exige incluir dígitos en el patrón.
LanguageRegionType	Patrón [a-z]{2}-[a-z0-9]{2}	El valor "es-e1" en 4.757 registros exige ampliar la parte de región.

07

Desarrollo del XSLT

Diseño, Estructura, Decisiones Técnicas y Aplicaciones

Generación del XML de Muestra

- 6 preguntas de Science & Mathematics.
- 6 preguntas de Education & Reference.
- 4 preguntas de Social Science.
- 6 preguntas de Arts & Humanities.

Criterio	Resultado
Total de preguntas	22
Con vot_date (votacion comunitaria)	7
Sin vot_date (elegida por el autor)	15
Sin campo content (elemento ausente en el XML)	2
Idiomas distintos del inglés	0 (dataset predominantemente en-us)

generar_muestra_xslt.py

↓

yahoo_muestra.xml

Diseño y Estrcutura del XSLT

yahoo_muestra.xml

↓

yahoo_answers.xslt

Template	Funcion
match="/"	Template raiz. Genera la estructura *HTML* completa: cabecera con estadísticas calculadas mediante *XPath*, botones de filtro por categoría, *grid* de tarjetas y *footer*.
match="vespaadd"	*Template* principal. Genera una tarjeta *HTML* por cada registro. Determina la clase de categoría, extrae todos los campos y gestiona los campos opcionales.
name="unix-to-date"	Convierte un Unix *timestamp* a fecha legible (ej: 25 Sep 2005) mediante el algoritmo de Howard Hinnant.
name="flag"	Devuelve el emoji de bandera correspondiente al código de región *qintl*.

Aplicación del XSLT y Resultado

aplicar_xslt.py

Yahoo! Answers

KNOWLEDGE ARCHIVE · OCTOBER 2007 · BBDD AVANZADAS · UCLM

Dataset: FullOct2007 · 4.483.032 records total Subset: 4 categories · 613.245 questions Sample: 22 questions displayed

22	6	6	4	6	7	2
TOTAL QUESTIONS	SCIENCE & MATH	EDUCATION & REF	SOCIAL SCIENCE	ARTS & HUMANITIES	COMMUNITY VOTED	NO CONTENT FIELD

FILTER BY CATEGORY: ▪ SCIENCE & MATH ▪ EDUCATION & REF ▪ SOCIAL SCIENCE ▪ ARTS & HUMANITIES

⚙ SCIENCE & MATHEMATICS — us en-us

↳ Biology

Why do people blush when they are embarrassed?

Why do people blush when they are embarrassed?

✍ BEST ANSWER · CHOSEN BY AUTHOR

from ask yahoo...

 http://ask.yahoo.com/ask/20040113.html

 <<Blushing is a unique blend of evolutionary and social behavior. It's an involuntary reaction of the sympathetic nervous system, which is responsible for our "fight or flight" response, but blushing is solely triggered by social cues.
 People generally blush when they're feeling embarrassed, scared, or stressed. As a result of the "fight or flight" response, the capillaries that carry blood to the skin widen, and the increased blood flow lends the face, as well as sometimes the chest, neck, or even the body or legs, a reddened color.

 Excessive facial blushing, or erythrophobia, is caused by overactivity of the sympathetic nervous system. The condition can cause a lot of psychological duress and has engendered several support groups.

 It's common knowledge that animals don't blush. So while there are some evolutionary cues behind blushing, it's also linked to something uniquely human -- moral consciousness.>>

1 alternative answer(s) — click to expand

Posted: 7 Jul 2005 · Resolved: 10 Aug 2005 · Author pick · u1254780 · #4189466

⚙ SCIENCE & MATHEMATICS — us en-us

↳ Biology

How the human species evolved?

How the human species evolved?

🏆 BEST ANSWER · VOTED BY COMMUNITY

A tough question as it overlaps science and theology. Since you asked "how the human species evolved?" I'll assume you're interested in the scientific approach to the answer.

 The current theory holds that Homo sapiens evolved from Hominid ancestors over the course of millions of years through a process called natural selection. Natural selection is the weeding and advancement of species variants over time based on the fit in that species' then current environments. The pressures for "fit" are competitive (with each other, with other tribes, and with other animals) and environmental (weather, terrain, availability of food, etc.). These factors all contributed to the development of our higher brain, which we now use to evolve as a global collective through communication & technology.

2 alternative answer(s) — click to expand

Posted: 25 Sep 2005 · Resolved: 4 Oct 2005 · Community pick · u143603 · #1251537

⚙ SCIENCE & MATHEMATICS — us en-us

↳ Mathematics

What is an "imaginary number"?

What is an "imaginary number", and how is it treated in algebra equations?

✍ BEST ANSWER · CHOSEN BY AUTHOR

Imaginary numbers are numbers than when squared equal a negative number, as in i^2 = -1, where i is the imaginary number. You'll also often see them represented as i = √-1 (that's the square root of -1).

 Don't be confused by the poorly chosen name - imaginary numbers do indeed exist and are used in advanced math, such as in the physics of electromagnetic fields. The analogy that Wikipedia uses is a good one - just like you don't need the concept of fractions to count stones, it doesn't mean that fractions don't exist. :)

1 alternative answer(s) — click to expand

Posted: 25 Aug 2005 · Resolved: 26 Aug 2005 · Author pick · u1372188 · #1568482

⚙ SCIENCE & MATHEMATICS — us en-us

⚙ SCIENCE & MATHEMATICS — us en-us

⚙ SCIENCE & MATHEMATICS — us en-us

EDUCATION & REFERENCE

us en-us

↳ Trivia

How to get rid of a beehive?

> *Who can tell me how to get rid of a beehive with out getting stung?*

BEST ANSWER · CHOSEN BY AUTHOR

Call an area apiarist. They should be able to help you and would most likely remove them at no charge in exchange for the hive. The bees have value and they now belong to you.

2 alternative answer(s) — click to expand

Posted: 16 Aug 2005
Resolved: 18 Aug 2005

Author pick | u1233931 | #2874684

EDUCATION & REFERENCE

us en-us

↳ Primary & Secondary Education

Which are the best K-12 public schools in the SF Bay area?

> *Which are the best K-12 public schools in the SF Bay area?*

BEST ANSWER · VOTED BY COMMUNITY

it depends on what you're looking for.

 Cupertino district has some of the best test scores, but the flavor of the classes and homework is very intense. creativity is not promoted, but instead repetitive training and exercises are encouraged. Some people find this stifling, but this district is VERY popular especially with people of asian origin.

 Neighboring Saratoga has similar or better scores, but the corriculum encourages much more diversity and creativity.

 Palo Alto has a mixture of very good and average schools, so it depends which ones you get in to. The flavor is generally similar to Saratoga, and maybe a tad less competitive.

 Another district to consider is Los Altos, the main problem there is that the high school is not very good.

 The attached URL will give you very good insight into the scores and quality of the schools as well as demographics, but unfortunately very little insight into the "flavor" of each.

4 alternative answer(s) — click to expand

Posted: 25 Sep 2005
Resolved: 4 Oct 2005

Community pick | u241843 | #663210

EDUCATION & REFERENCE

us en-us

↳ Words & Wordplay

What's the longest english word without a vowel in it?

> *and what does that word mean if it's not a common word.*

BEST ANSWER · CHOSEN BY AUTHOR

The longest word without a vowel is Rhythm.

 It is referenced on below web site where you can find more fun facts.

2 alternative answer(s) — click to expand

Posted: 25 Sep 2005
Resolved: 29 Sep 2005

Author pick | u23481 | #2542068

SOCIAL SCIENCE

us en-us

↳ Psychology

What is the difference between Askers and Answerers?

> *Is it the same difference as Talkers vs Doers, Dumbos vs Smarties? Do people tend to fall primarily into one category or the other?*

BEST ANSWER · VOTED BY COMMUNITY

one can argue that in order to become an answerer you first need to be an asker- how would you gather a more then common knowledge information- you need to inquire, thus ask many questions. only then you can answer other's people questions. So- He who wants to become an answerer must first be an asker

SOCIAL SCIENCE

us en-us

↳ Psychology

I need tips to be a strong leader.?

BEST ANSWER · CHOSEN BY AUTHOR

Never show weakness..always think things through..lead by example...be curtious of your followers...and learn by your mistakes.

9 alternative answer(s) — click to expand

SOCIAL SCIENCE

us en-us

↳ Gender & Women's Studies

Do you Believe in God? Why? Has he ever helped you?

BEST ANSWER · CHOSEN BY AUTHOR

I opened my eys to the world. There's no god. I wake uo myslef.

2 alternative answer(s) — click to expand

07

Resultados

Consultas, Visualización, y Análisis

Alberto Lillo García · Manuel Caballero Bonilla
Bases de Datos Avanzadas · UCLM - ESI · Marzo 2026

613.245 PREGUNTAS · 4 CATEGORÍAS · 5 BLOQUES DE ANÁLISIS XQUERY

SCIENCE & MATHEMATICS
183.354
29,90% · 65 subcategorías

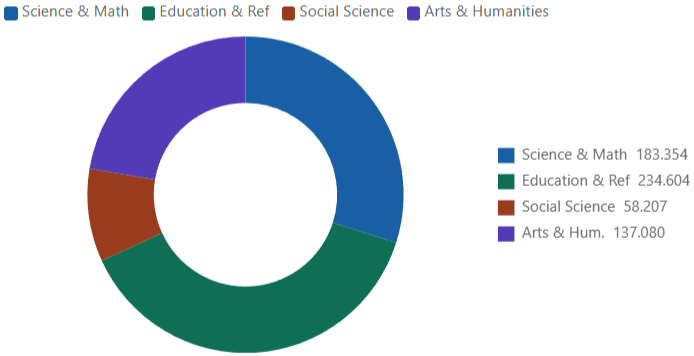
EDUCATION & REFERENCE
234.604
38,26% · 103 subcategorías

SOCIAL SCIENCE
58.207
9,49% · 33 subcategorías

ARTS & HUMANITIES
137.080
22,35% · 54 subcategorías

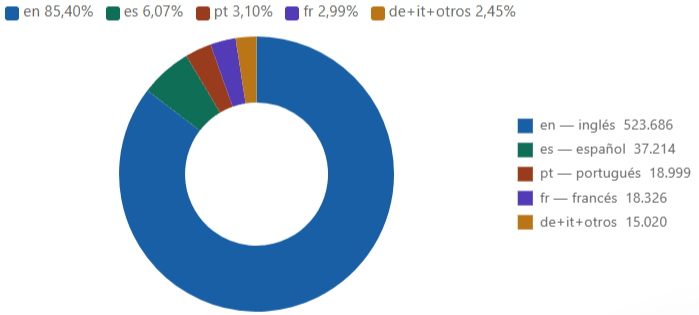
Distribución del subset extraído

4.483.032 registros en el dataset original · 178 categorías disponibles · seleccionadas 4



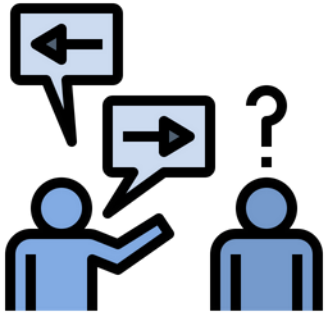
Idiomas del subset (campo qlang)

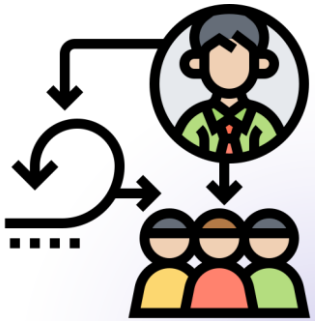
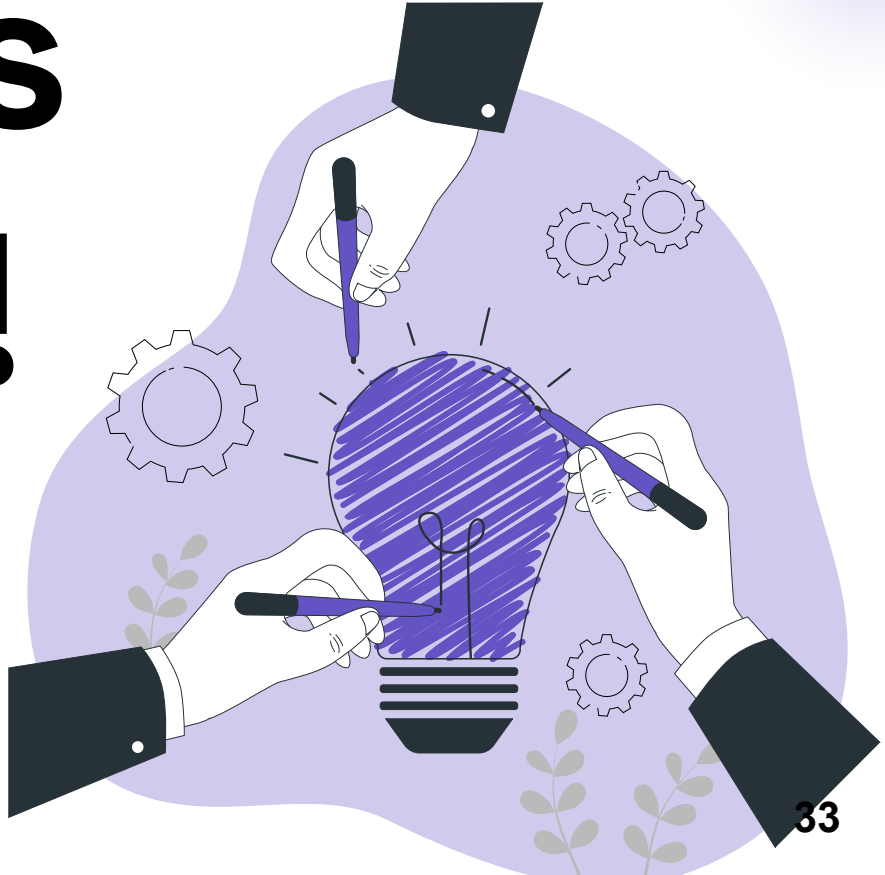
16 códigos de idioma distintos · el inglés representa más del 85% del total



Nota: El 6,07% en español incluye el código de región "e1" (Yahoo interno, no estándar ISO), que explica por qué combinaciones como "es-e1" aparecen en el dataset. Se analiza en detalle en el Bloque 4.

CONCLUSIONES





¡¡¡Muchas Gracias!!!